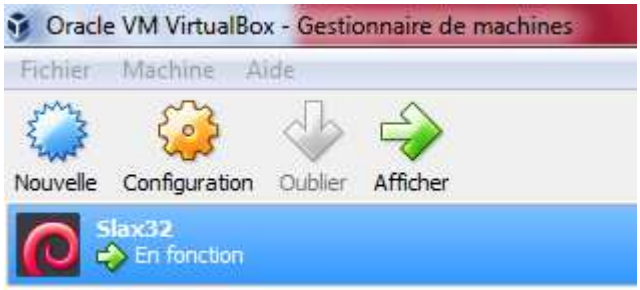
TP08 PAS A PAS (Utilisation de Buildroot)

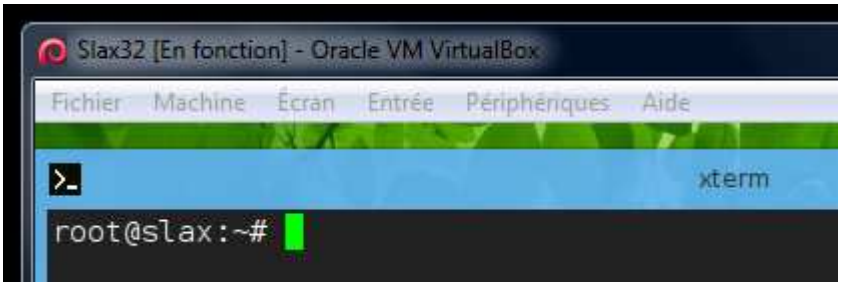
Matière : ATELIER SYSTEME D'EXPLOITATION EMBARQUE | Classe : SEM21

Travail effectué

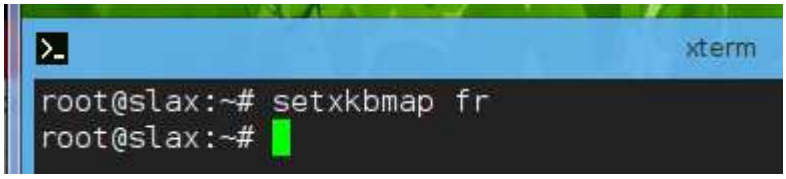
1- Lancer VirtuelBox



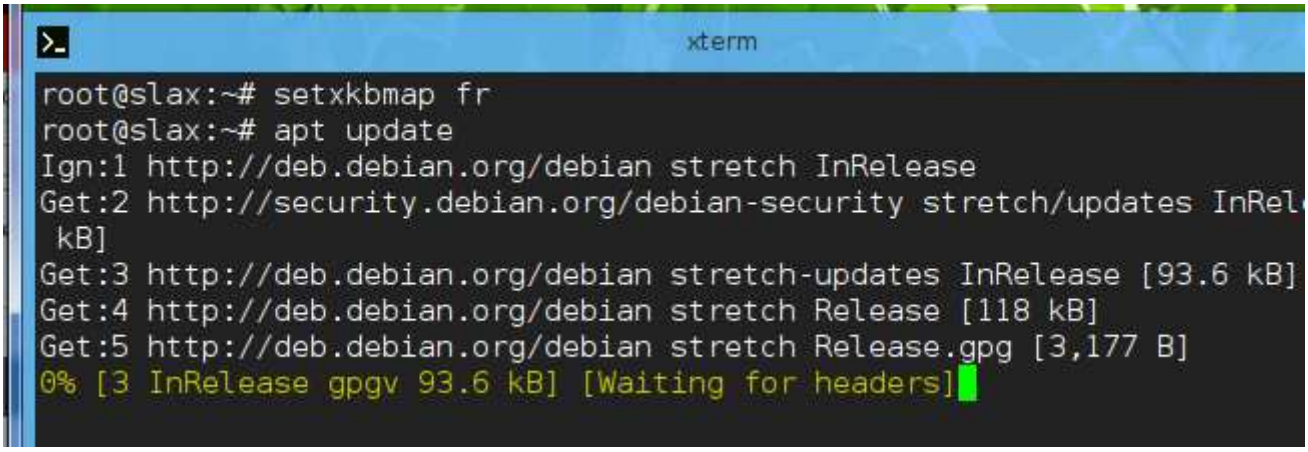
2- Démarrer la machine virtuelle "Slax32"



3- Changer le clavier en "azerty"



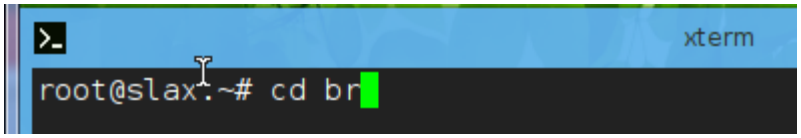
4- Actualiser apt (apt update)



5- Créer un dossier "br" (mkdir)



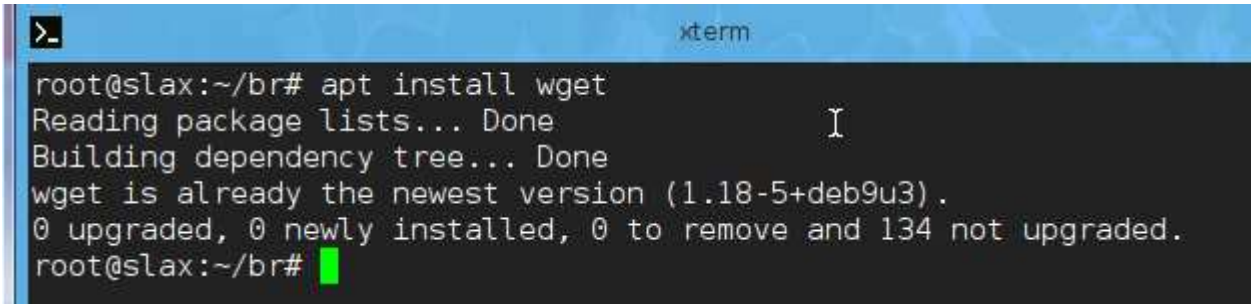
6- Entrer dans le dossier "br" (cd)



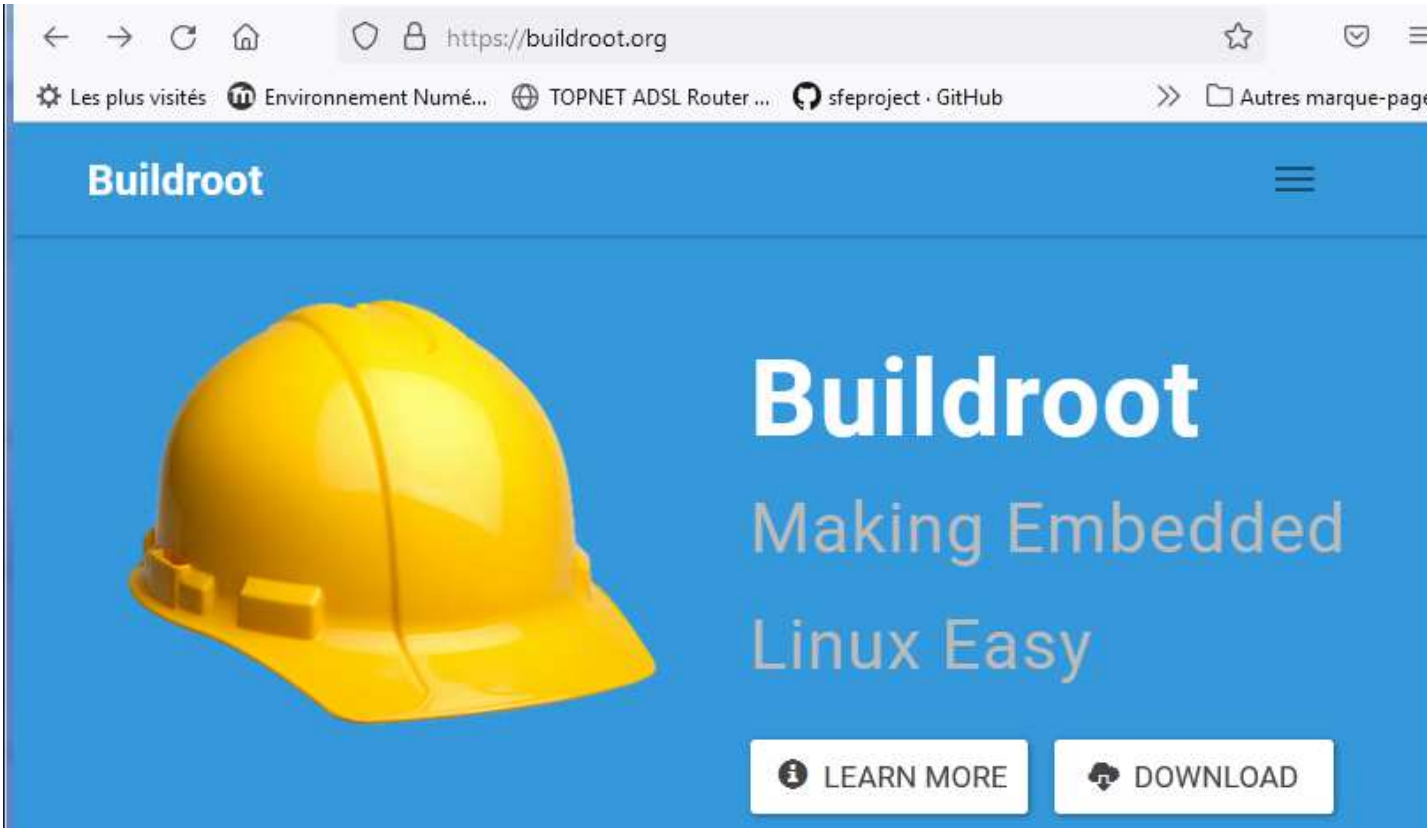
7- Lister le contenu de "br" (ls)



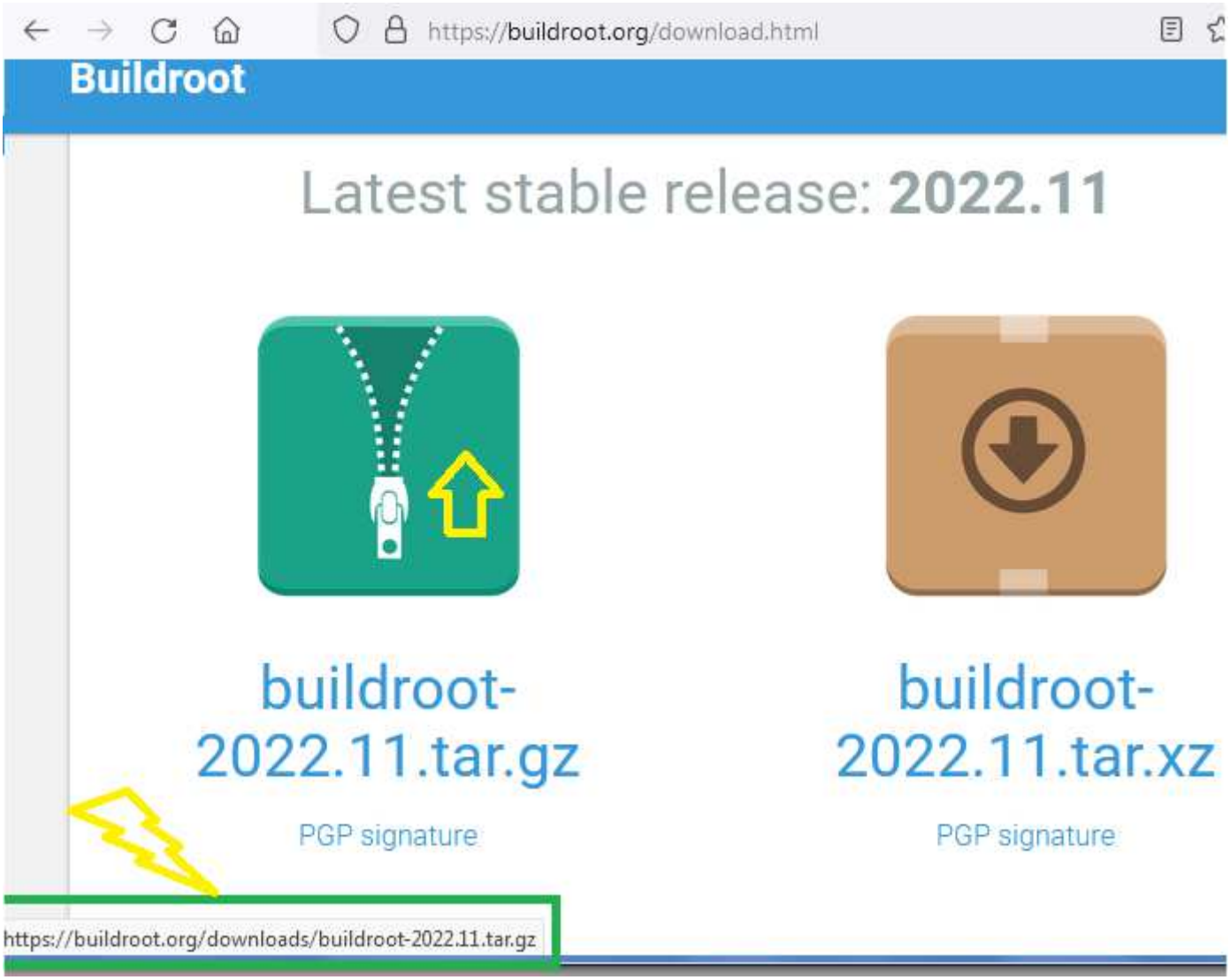
8- Installer wget



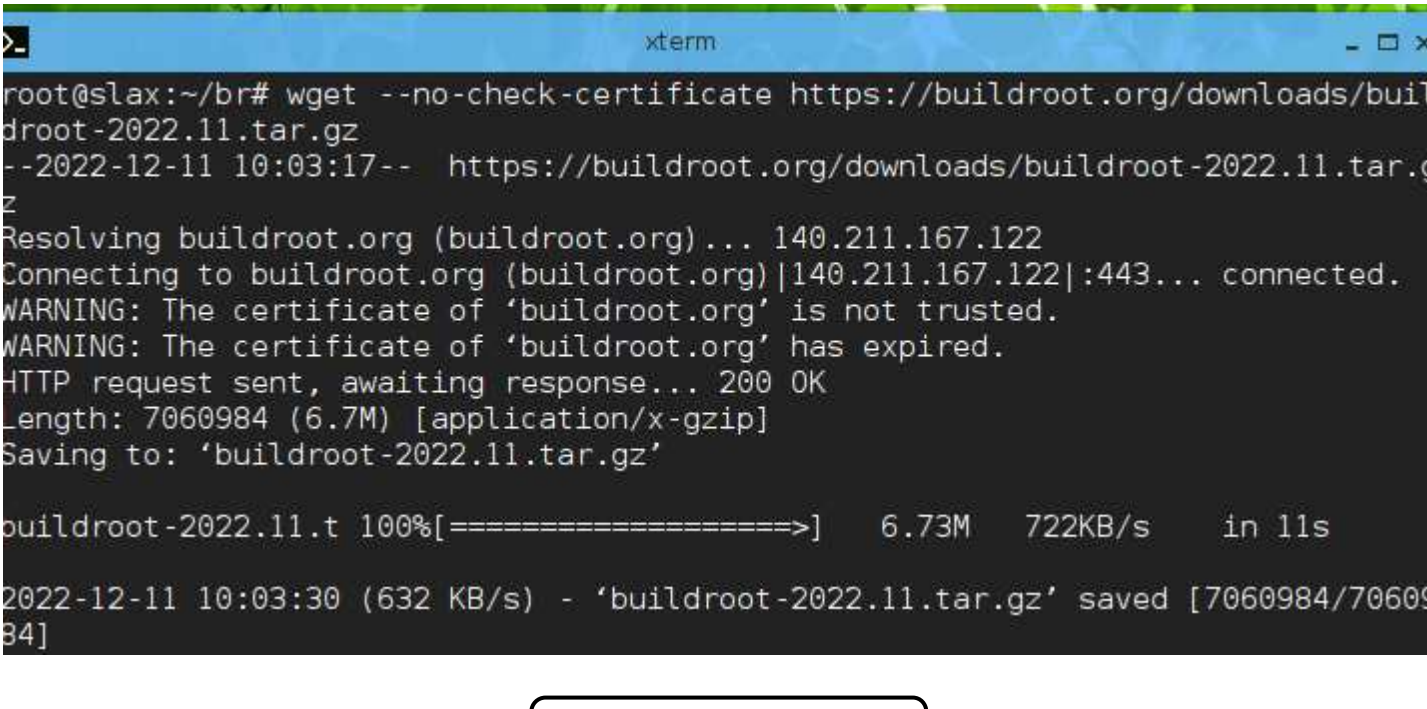
9- Ouvrir le lien [1] et cliquer sur "Download"



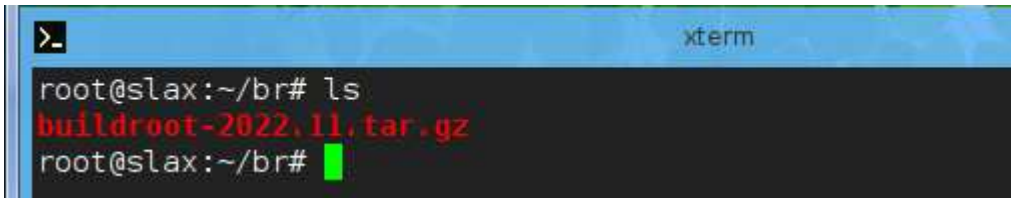
10- Fixer la souris sur "buildroot-2020.1.11.tar.gz" pour voir son chemin complet dans la barre d'état



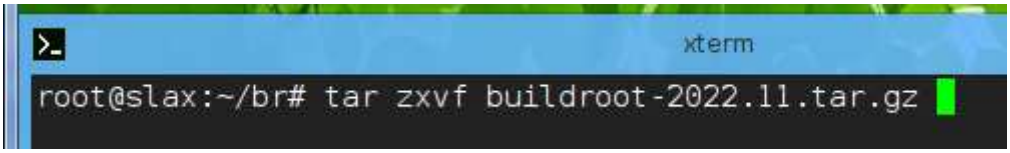
11- Télécharger "buildroot-2020.11.1.tar.gz" (wget --no-check-certificate CheminComplet)



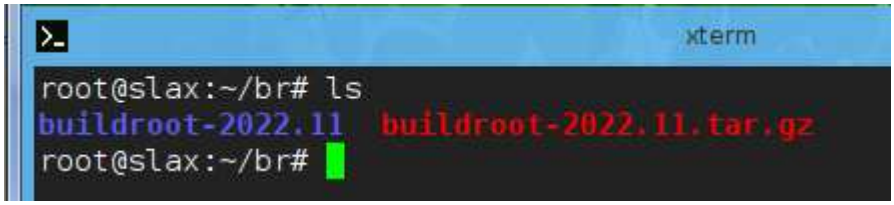
12- Lister le contenu du dossier en cours (ls)



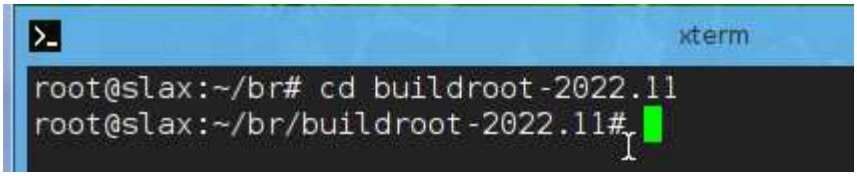
13- Extraire le fichier "buildroot-2020.1.11.tar.gz" (tar zxvf nomFichier)



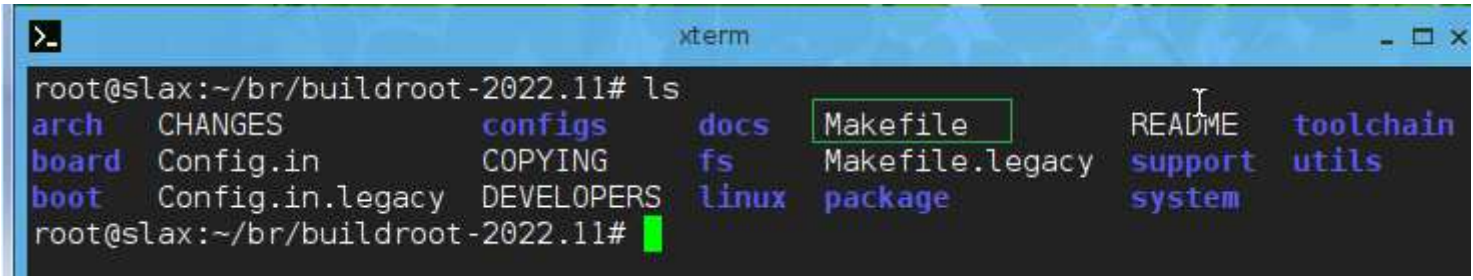
14- Lister le contenu du dossier en cours (ls)



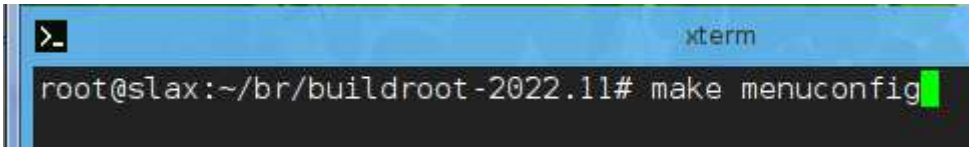
15- Entrer dans le dossier " buildroot-2020.1.11" (cd)

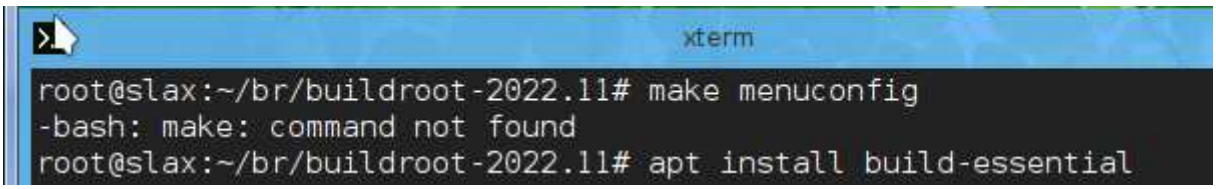


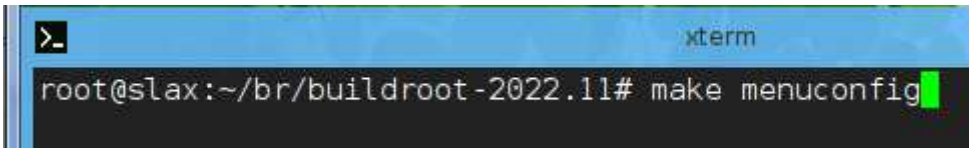
16- Lister le contenu du dossier en cours (ls)



17- Afficher le menu de configuration de buildroot (make menuconfig)

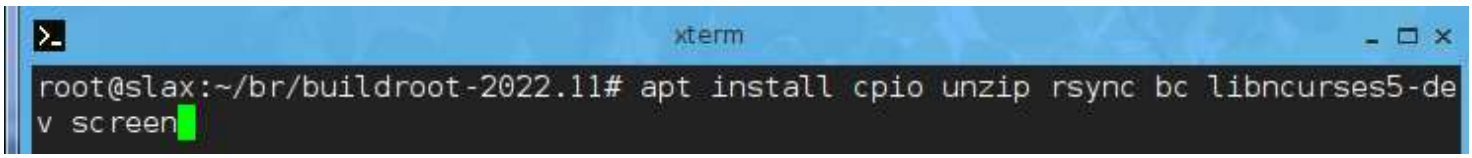




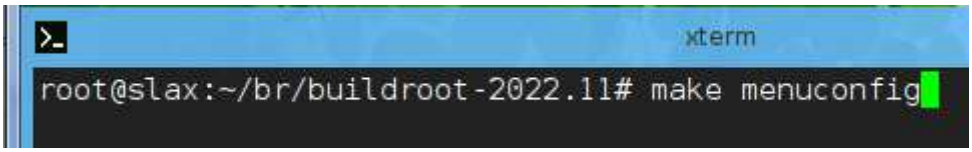


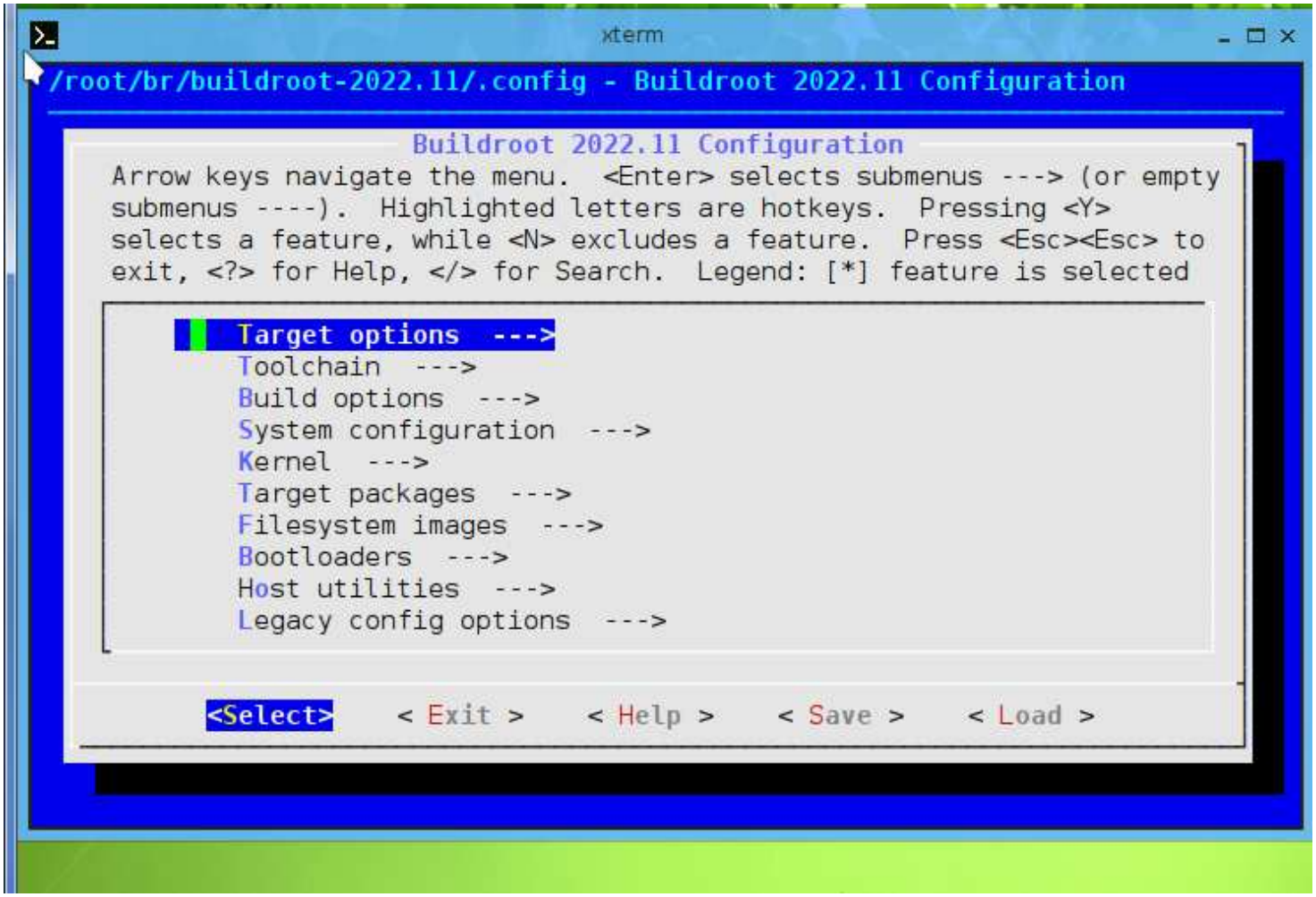

```
 *** Unable to find the ncurses libraries or the
 *** required header files.
 *** 'make menuconfig' requires the ncurses libraries
 ***
 *** Install ncurses (ncurses-devel or libncurses-dev
 *** depending on your distribution) and try again.
 ***
Makefile:253: recipe for target '/root/br/buildroot-2022.11/output/build/buildro
ot-config/dochecklxdialog' failed
make[1]: *** [/root/br/buildroot-2022.11/output/build/buildroot-config/dochecklx
dialog] Error 1
make[1]: Leaving directory '/root/br/buildroot-2022.11/support/kconfig'
Makefile:957: recipe for target '/root/br/buildroot-2022.11/output/build/buildro
ot-config/mconf' failed
make: *** [/root/br/buildroot-2022.11/output/build/buildroot-config/mconf] Error
 2
root@slax:~/br/buildroot-2022.11#
```

18- Installer toutes les bibliothèques du lien [2]



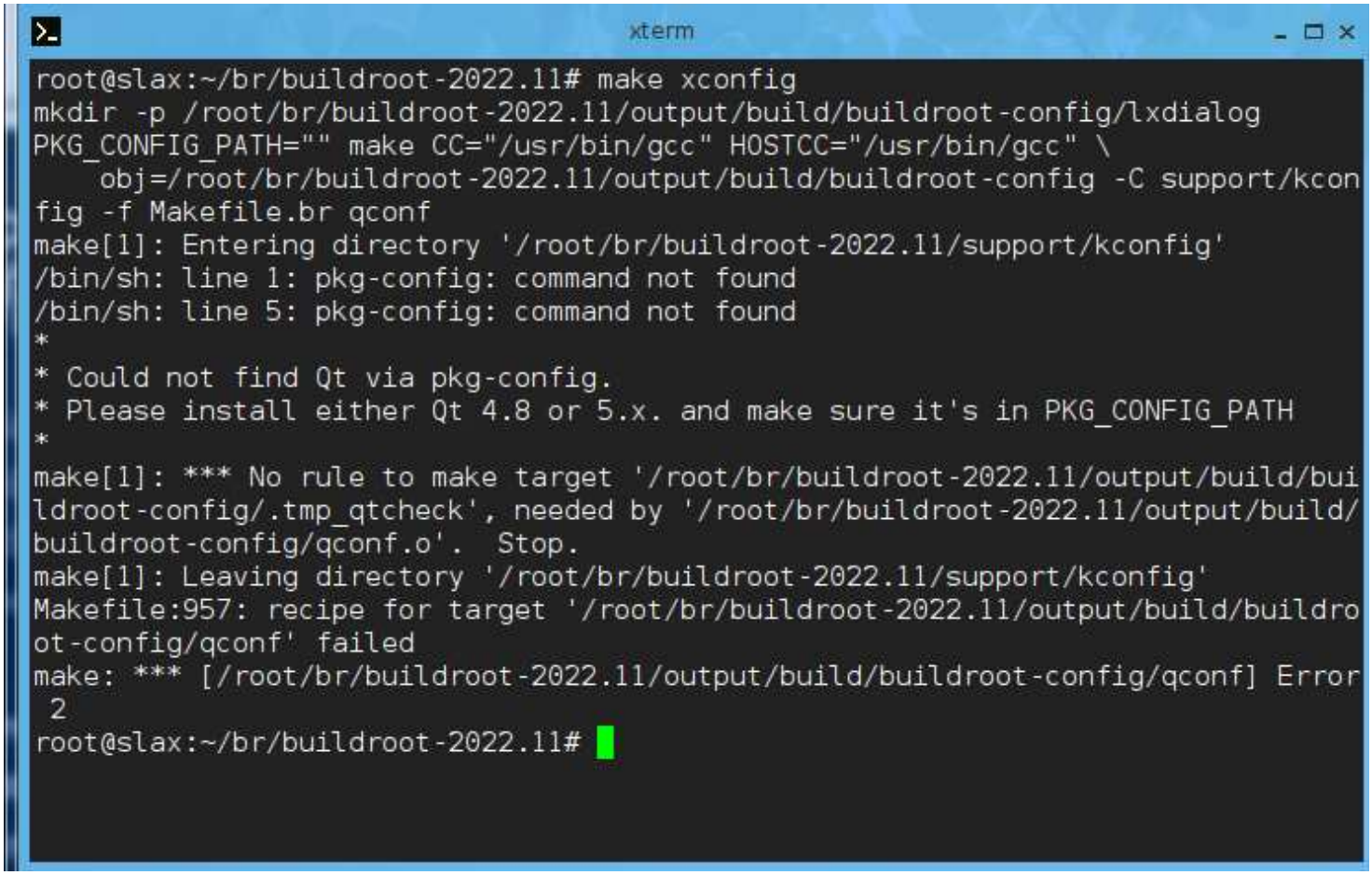
19- Afficher le menu de configuration de buildroot (make menuconfig)





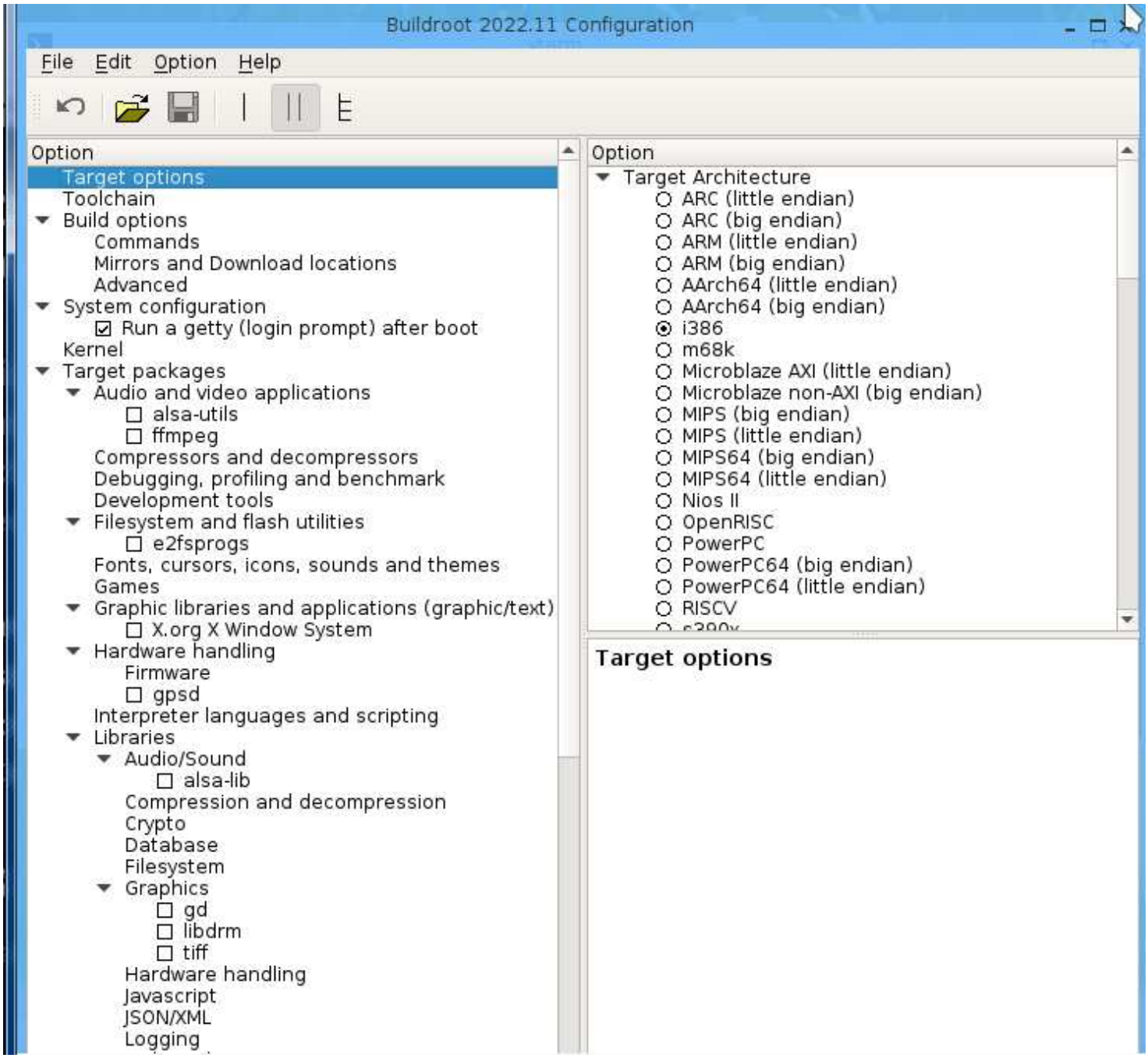
20- Quitter le "menuconfig"

21- Afficher le menu de configuration "qt" de buildroot (make xconfig)

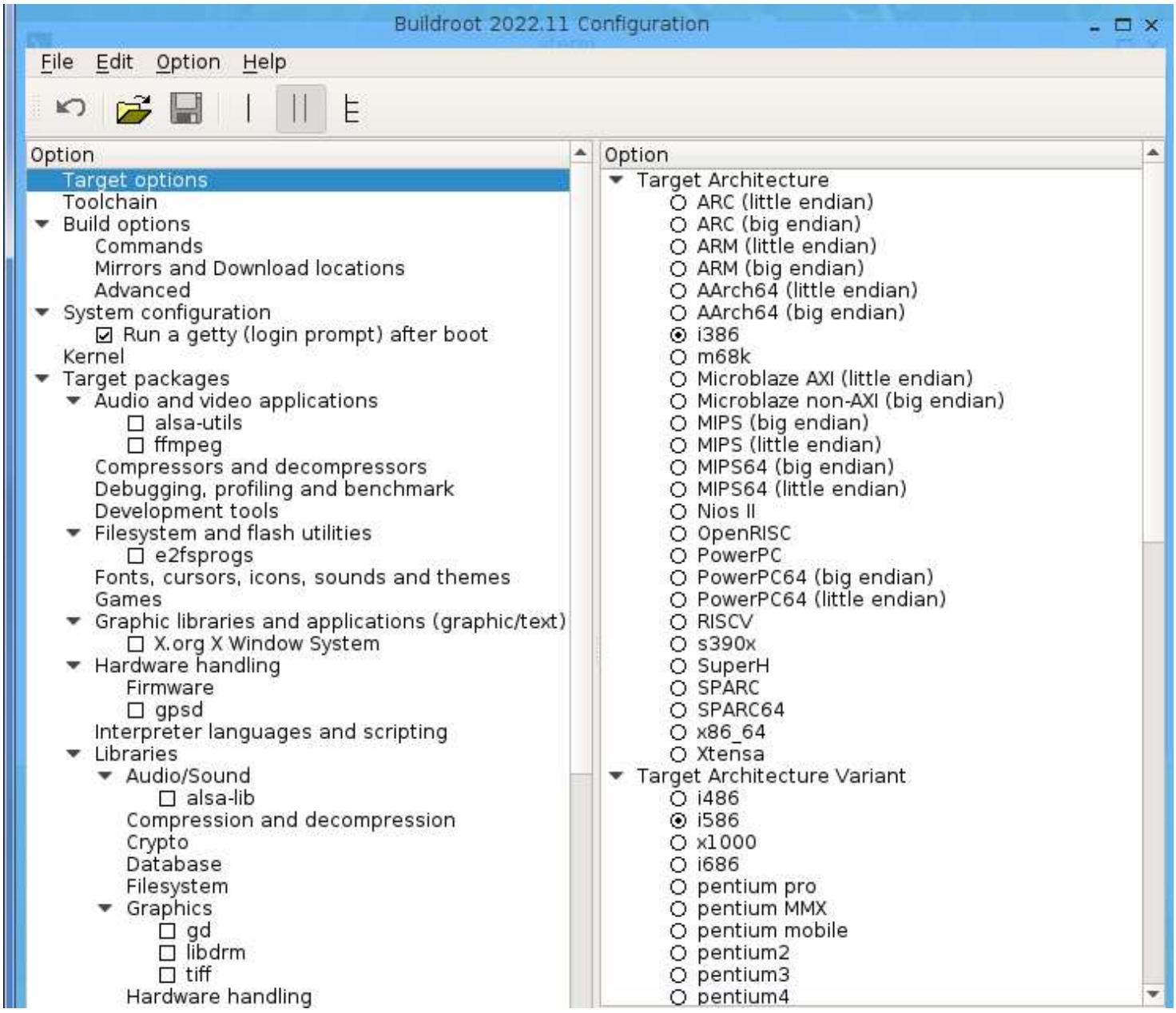


22- Installer Qt : apt install qtbase5-dev qtchooser qt5-qmake qtbase5-dev-tools qtcreator pkg-config -y

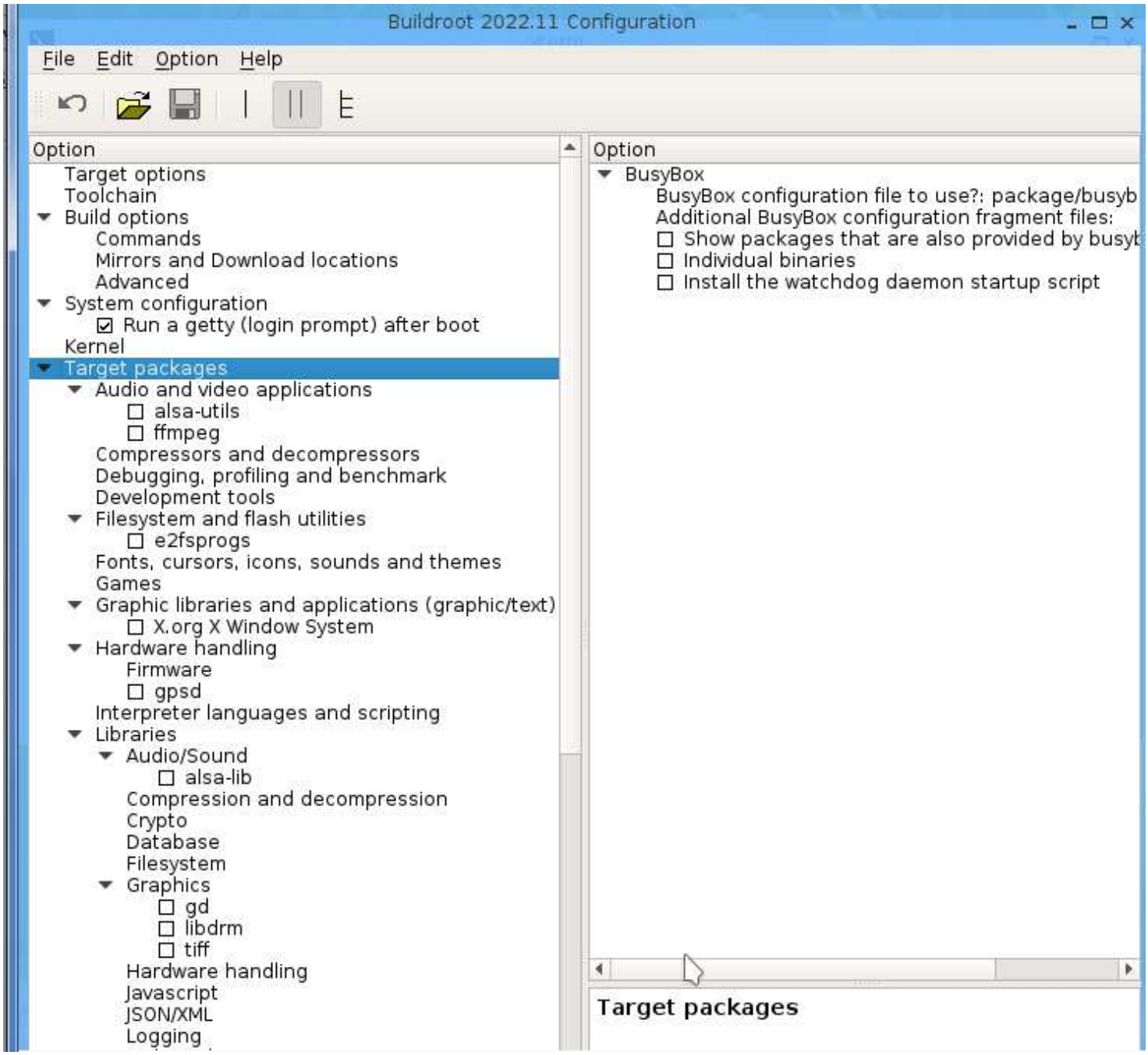
23- Afficher le menu de configuration "qt" de buildroot (make xconfig)


Buildroot 2022.11 Configuration
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e2fsprogs
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X.org X Window System
Hardware handling
Firmware
gpsd
Interpreter languages and scripting
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Crypto
Database
Filesystem
Graphics
gd
libdrm
tiff
Hardware handling
Javascript
JSON/XML
Logging
Option
Target Architecture
ARC (little endian)
ARC (big endian)
ARM (little endian)
ARM (big endian)
AArch64 (little endian)
AArch64 (big endian)
i386
m68k
Microblaze AXI (little endian)
Microblaze non-AXI (big endian)
MIPS (big endian)
MIPS (little endian)
MIPS64 (big endian)
MIPS64 (little endian)
Nios II
OpenRISC
PowerPC
PowerPC64 (big endian)
PowerPC64 (little endian)
RISCV
Target options

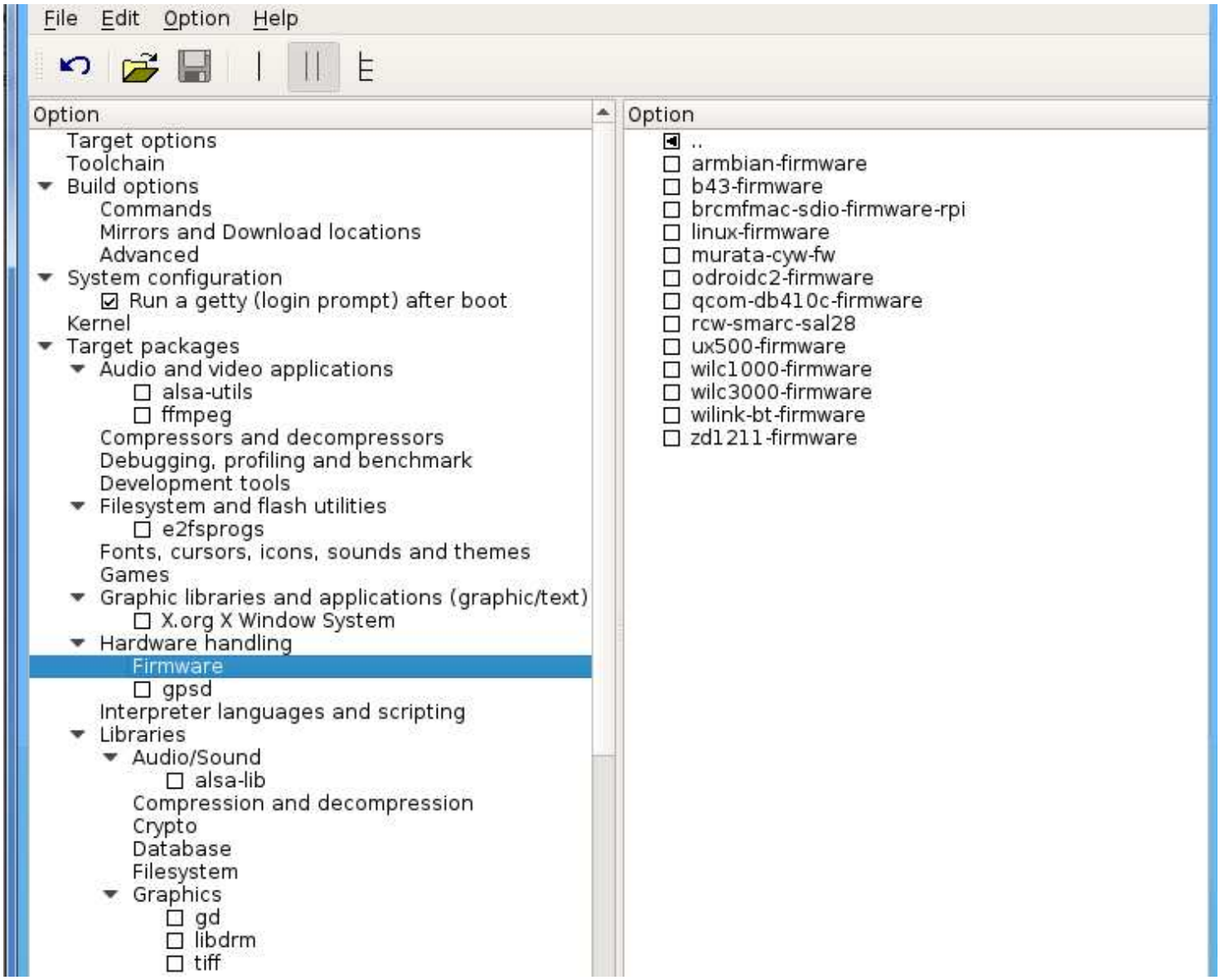
Target options



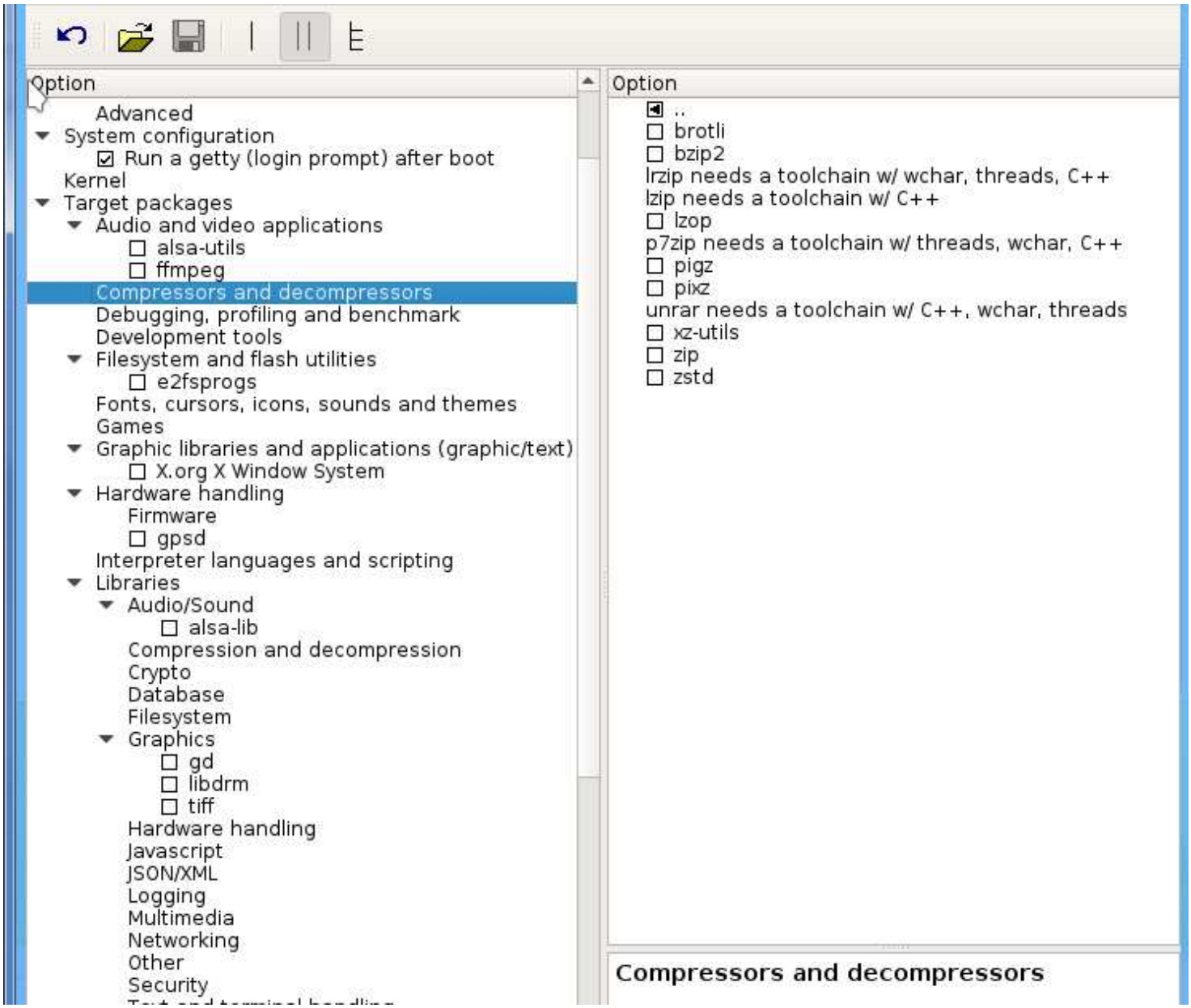
Target packages



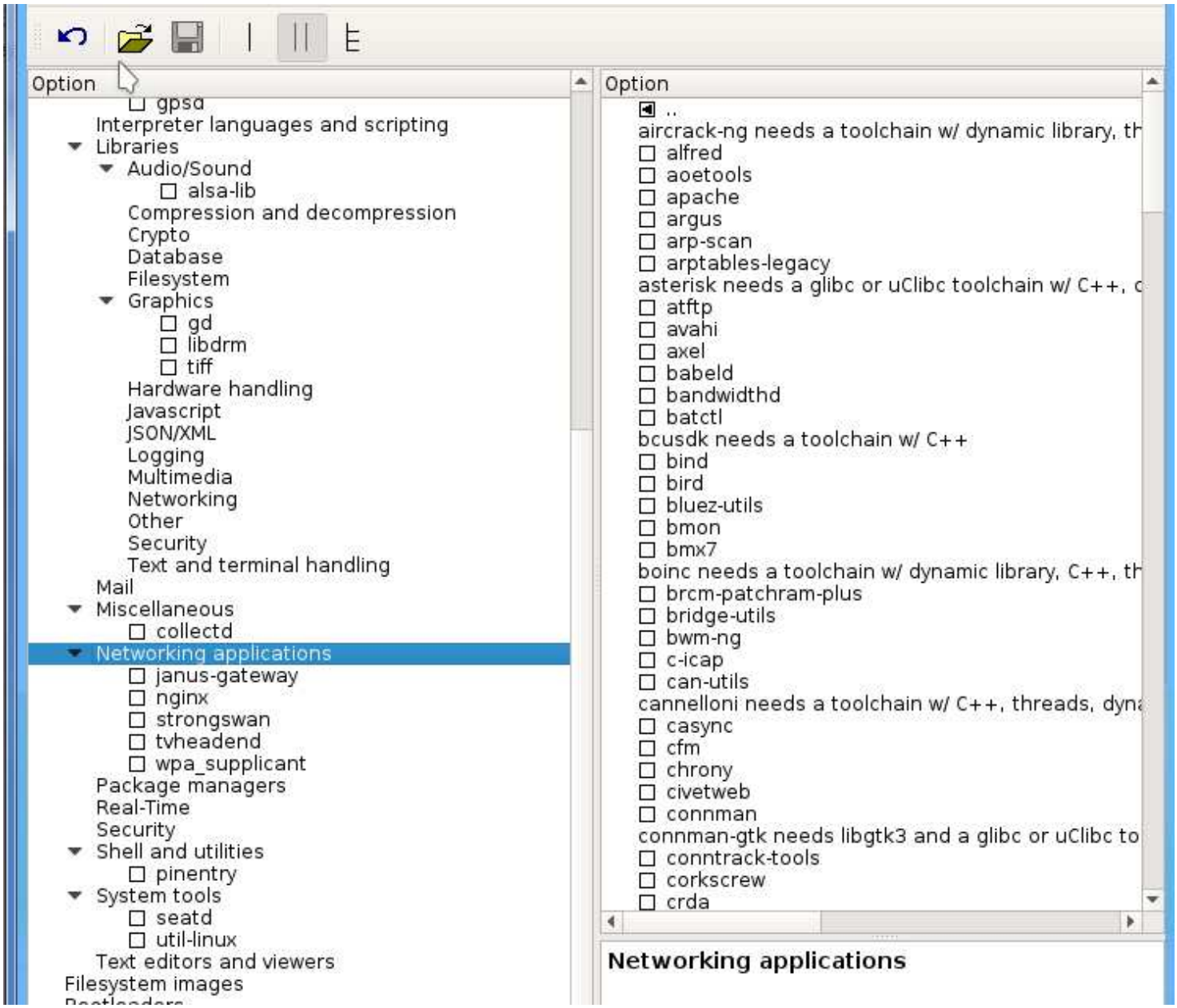
Firmware

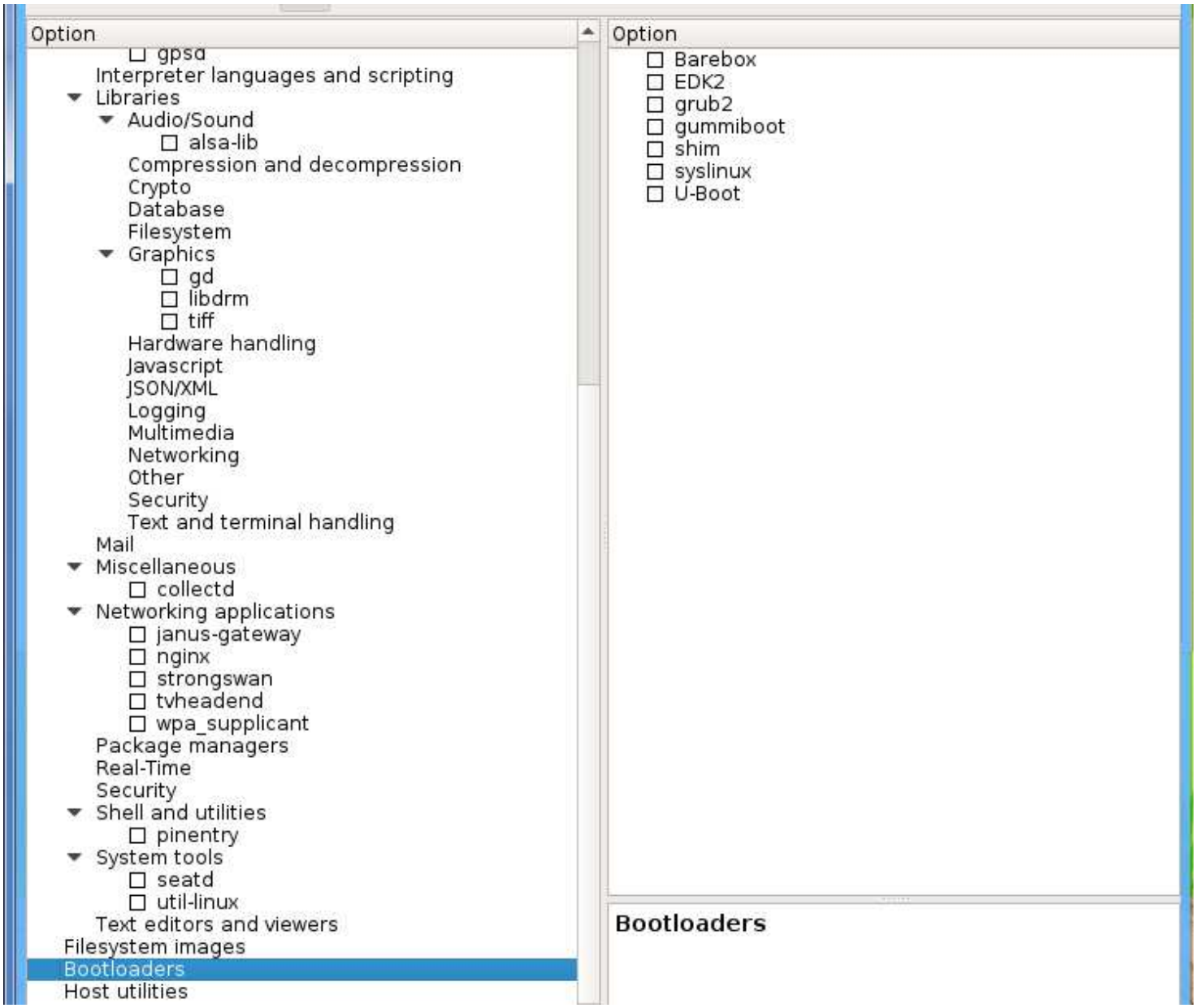

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brcmfmac-sdio-firmware-rpi
linux-firmware
murata-cyw-fw
odroidc2-firmware
qcom-db410c-firmware
rcw-smarc-sal28
ux500-firmware
wilc1000-firmware
wilc3000-firmware
wilink-bt-firmware
zd1211-firmware

Compressors and décompressors

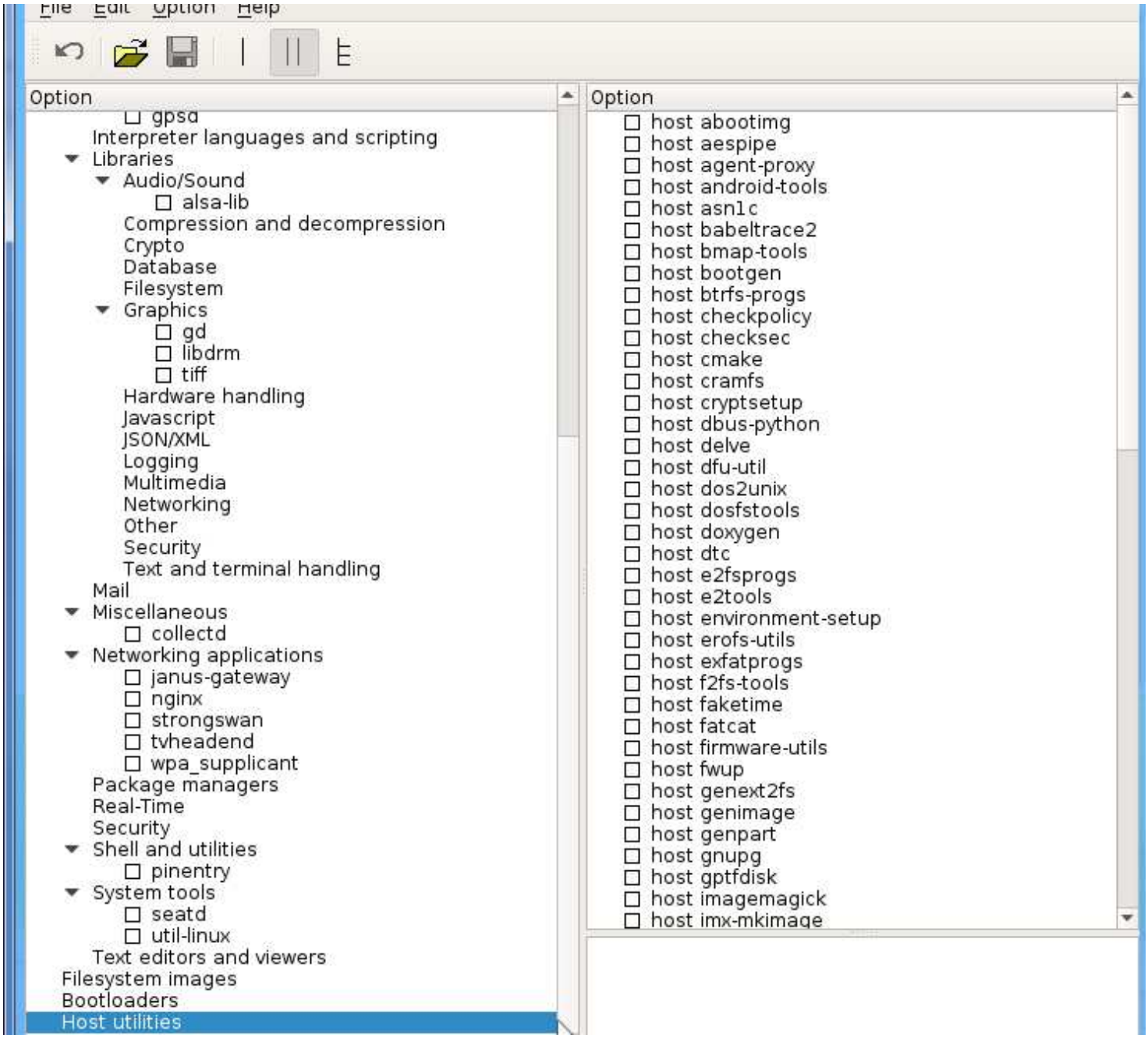


Networking applications

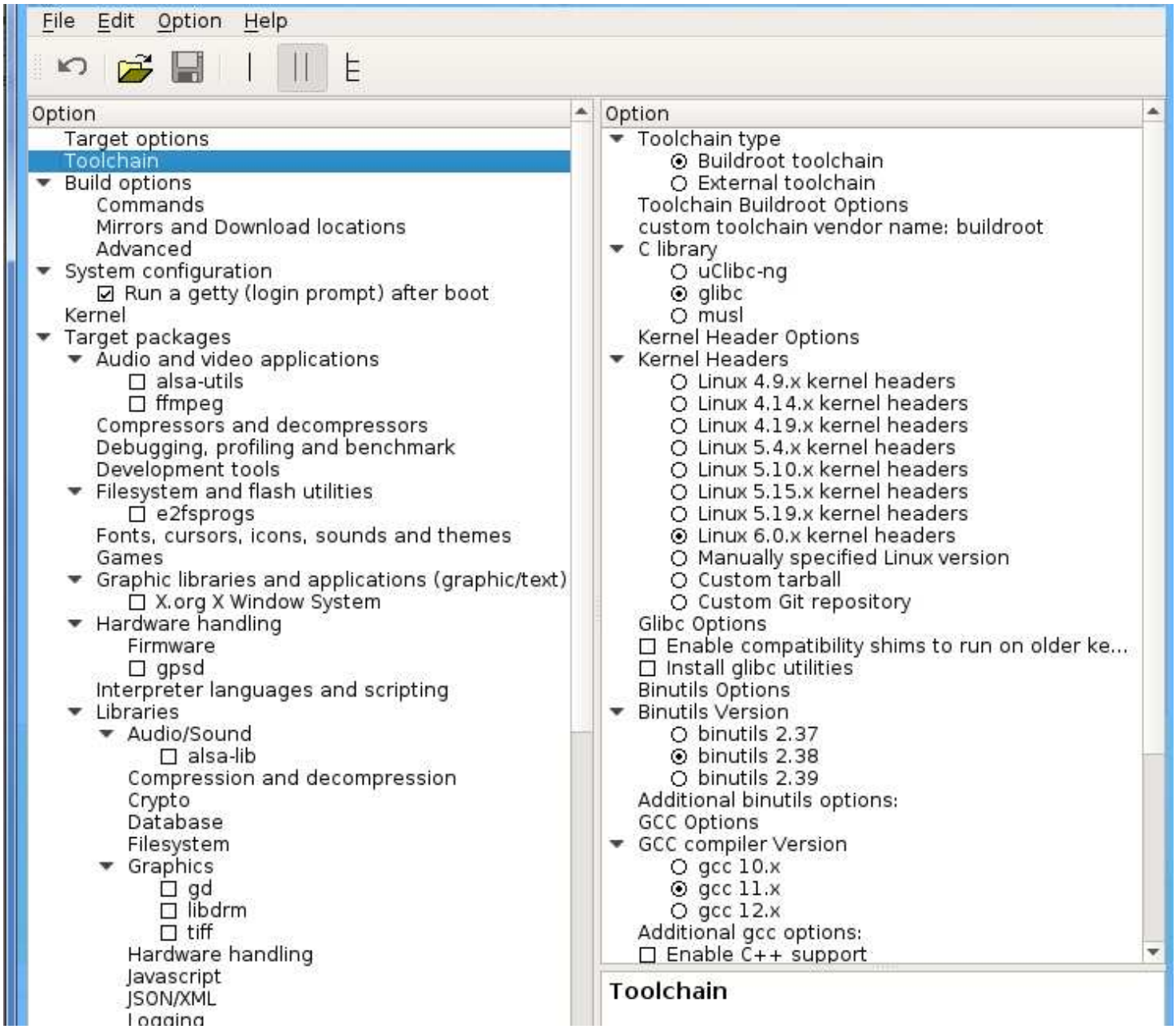



Option
gpsd
Interpreter languages and scripting
Libraries
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alsa-lib
Compression and decompression
Crypto
Database
Filesystem
Graphics
gd
libdrm
tiff
Hardware handling
Javascript
JSON/XML
Logging
Multimedia
Networking
Other
Security
Text and terminal handling
Mail
Miscellaneous
collectd
Networking applications
janus-gateway
nginx
strongswan
tvheadend
wpa_supplicant
Package managers
Real-Time
Security
Shell and utilities
pinentry
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seatd
util-linux
Text editors and viewers
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Option
Barebox
EDK2
grub2
gummiboot
shim
syslinux
U-Boot
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Host utilities



Toolchain


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Toolchain Buildroot Options
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uClibc-ng
glibc
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Linux 4.14.x kernel headers
Linux 4.19.x kernel headers
Linux 5.4.x kernel headers
Linux 5.10.x kernel headers
Linux 5.15.x kernel headers
Linux 5.19.x kernel headers
Linux 6.0.x kernel headers
Manually specified Linux version
Custom tarball
Custom Git repository
Glibc Options
Enable compatibility shims to run on older ke...
Install glibc utilities
Binutils Options
Binutils Version
binutils 2.37
binutils 2.38
binutils 2.39
Additional binutils options:
GCC Options
GCC compiler Version
gcc 10.x
gcc 11.x
gcc 12.x
Additional gcc options:
Enable C++ support
Toolchain